

UTKARSH SHARMA

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Education

Indian Institute of Science (IISc)

Bangalore, India

M.Tech in Computer Science and Engineering | **CGPA: 8.7/10.0**

2024 – 2026

Key Coursework: Deep Learning for CV, Learning for 3D Vision and Inverse Graphics, Graphics & Visualization, Computational Methods of Optimization, Machine Learning for Signal Processing, Design and Analysis of Algorithms

Chandigarh University

Mohali, India

B.E. in Computer Science | **CGPA: 8.24/10.0**

2020 – 2024

Technical Skills

Languages: Python, C++, CUDA, SQL

Computer Vision: Physically Based Rendering (PBR), Neural Path Guiding, Neural Rendering, NeRFs, 3DGS

Deep Learning: Vision Transformers, Autoencoders, Normalizing Flows, Diffusion Models, Self-Supervised Learning

Frameworks & Tools: PyTorch, NumPy, Mitsuba 3, PBRT, Tiny-CUDA-NN, Dr.Jit, Pandas, Git

Research & Projects

Neural Path Guiding for Path Tracing | Mitsuba 3, PyTorch, Tiny-CUDA-NN

June – Dec 2025

- Implemented path guiding techniques like **Neural Parametric Mixtures (NPM)**, **Neural Importance sampling (NIS)**, **Distribution Factorization (DF)** for guiding the ray direction in Path Tracing reducing the variance by $2 - 4\times$ compared to BRDF sampling.
- Implemented **Neural Radiance Caching** for real-time global illumination approximation, using hierarchical spatial hash encoding and online gradient-based optimization during the render loop.

CUDA Path Tracer with Monte Carlo Radiosity | C++, CUDA, Python

Aug – Dec 2025

- Developed a renderer containing both unbiased **Path Tracing** and **Radiosity** for physically accurate diffuse inter-reflections and caustic light transport.
- Implemented custom **Bounding Volume Hierarchy (BVH)** for faster ray intersection speed.
- Integrated multiple importance sampling (MIS) for combining BSDF and light sampling strategies, reducing fireflies and noise in difficult lighting conditions.

Masked Autoencoder (MAE) using Vision Transformer (ViT) | PyTorch, Self-Supervised Learning

Oct – Dec 2025

- Implemented a Masked Autoencoder with a transformer encoder-decoder architecture.
- Trained the model to reconstruct missing image patches from latent representations, enabling self-supervised pre-training
- Visualized attention maps to interpret how the model aggregates context from unmasked patches to predict occlusion.

Deep Learning-Based Denoising for Monte Carlo Rendering | PyTorch, PBRT

Feb – Apr 2025

- Developed and benchmarked **diffusion-based and U-Net denoisers** for low-sample Monte Carlo path tracing outputs
- Incorporated **G-Buffer auxiliary features** (albedo, normals, depth, motion vectors) as multi-channel inputs to guide network predictions and preserve geometric edges and material boundaries.
- Achieved superior perceptual quality (SSIM > 0.95, PSNR > 35 dB) on PBRT-generated test scenes

Neural Radiance Fields & 3D Gaussian Splatting | PyTorch

Aug – Dec 2025

- Built a **NeRF pipeline from scratch** using PyTorch, implementing volumetric rendering equations, sinusoidal positional encoding, and hierarchical stratified sampling to reconstruct complex 3D scenes.
- Implemented **3D Gaussian Splatting**, optimizing 3D covariance matrices and spherical harmonics (SH) coefficients.
- Achieved high-fidelity reconstruction on the **NeRF Synthetic dataset** (PSNR > 25 dB), validating the results.

Experience

Teaching Assistant – Algorithms & Programming | Indian Institute of Science

Aug 2025 – Dec 2025

- Facilitating weekly tutorial sessions for undergraduate students covering algorithm design, complexity analysis, and data structures.

Academic Achievements

Rank 8 out of 123,967 candidates in GATE Computer Science (CS) 2024

Rank 17 out of 39,210 candidates in GATE Data Science & AI (DA) 2024

Winner, CodeSmash 2.0 Competitive Programming Contest

Finalist, CodeRush 3.0 Competitive Programming Contest